INCH-POUND

MIL-K-71129 (AR)
AMENDMENT 4
17 December 1996
SUPERSEDING
AMENDMENT 3
23 April 1996

MILITARY SPECIFICATION

KIT, REFURBISHMENT, FOR CARTRIDGE, 81MM
PRACTICE, SHORT RANGE, M880
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-K-71129 (AR), dated 26 May 1992, and is approved for use by the U.S. Army Armaments Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

3.3: Delete: "propelling charge", and substitute: "ignition cartridge".

Delete 3.6.1 in its entirety and substitute:

"3.6.1 Misfires. The cartridges shall not fail to fire from the weapon when they are initially dropped into it."

* Delete 3.6.2 in its entirety and substitute:

"3.6.2 Short rounds. Cartridges having a horizontal range less than 80 percent of the average horizontal range (calculated without the short rounds) will be classified as a short round."

Delete 3.6.4 in its entirety and substitute:

"3.6.4 Range and deflection precision. The cartridge shall have a horizontal range probable error (PEr) less than 15 meters, for all zones when calculated to the nearest one tenth (0.1) meter. The deflection probable error (PEd) shall be less than 15 meters for all zones when calculated to the nearest one tenth (0.1) meter."

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3.6.6: Add: "unobstructed", before the word "flight".

3.6.9 Add the following:

3.6.9 Dud plug function.

"3.6.9.1 Functioned round. All dud plug heads (regardless of stems) shall blow out of the projectile body upon fuze function."

"3.6.9.2 Inert round. No dud plug head (regardless of stems) shall exit the projectile body during ballistic flight or upon impact."

3.9 Add the following:

"3.9 Dud plugs."

"3.9.1 Configuration. Three dud plugs shall be present and properly inserted into the projectile body for the proving ground test."

"3.9.2 Dud plug push in - push out force. No dud plug shall push into the required test plate using a force greater than 40.0 pounds and shall not be capable of being pushed out of the required test plate using a force of less than 25.0 pounds at a temperature of 70 degrees F ± -3 degrees F when tested as specified in 4.5.4."

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* 4.3.3(b), Add after first sentence:

"Graze impact duds shall not be counted for score. These shall be recorded for information only and shall be x-rayed to document that the fuze was armed. Side impact duds shall be recorded for information only and x-rayed to document that the fuze was armed. If more than four side impacts occur, the first article sample represented by the sample shall be referred to the technical agency for disposition. In either of the above cases, if the x-ray reveals that the fuze did not arm, the fuze shall be considered to be a dud."

* Delete "c" in its entirety and substitute:

"c. One or more short rounds occur at +110 degrees F or +70 degrees F.

Two or more short rounds occur at 0 degrees F."

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Delete "g" and "h" in their entirety and substitute:

"g. If the weapon is damaged by abnormal wear of the interior or of the firing pin, or caused by excessive pressure."

"h. Two or more improper impact signatures occur."

Add the following:

"i. All three dud plug heads (regardless of stems) are missing from the projectile body in the inert fuze ballistic test."

"j. Two or more dud plug heads (regardless of stems) fail to exit projectile bodies (accumulatively, i.e., two remaining in a single projectile or one each remaining in two projectiles) in the live fuze ballistic tests (except in the case of a particular round that was an actual dud, in which case paragraph "i." above applies)."

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Table I: Proving ground test line, under Conformance Criteria:

Delete: "72", and substitute: "96".

Add the following:

"Dud Plug ...(dwg. 12961001)

Examination for defects ...30 ... 3.1 ... 4.4.2.11

Push in - push out test ... 30 ... 3.9.2 ... 4.5.4".

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4.4.1: Delete the word "interfix" from the text from lines a., b., c., and d.

Add the following:

"e. Dud plugs from one inspection lot that comply with 3.9.2."

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4.4.2.1, Major 105: Change call out:

from: "Length of large outside diameter section",

to: "Length between large outside diameter aft face to firing pin point".

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4.4.2.5: Delete Special A in its entirety.

Critical: Add the following:

"1 Fuze striker in armed position ... 100% ... 3.1 ... Visual".

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4.4.2.6: Delete Major 103 in its entirety.

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4.4.2.10: Delete Specials A, B, C, D, E, F and G in their entirety and add the following:

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
“1	Fuze striker in armed position	(a)	3.1	Gage
2	Breech plug assembly firing pin not below flush after assembly to round	(a)	3.3	Gage
3	Powder from energetic cartridges found on outside of round	(a)	3.1	Visual
Major				
105	Red marking visible on fuze striker	(a)	3.4	Visual
106	Fuze packing clip missing or improperly assembled	(a)	3.1	Visual
107	Obturator ring missing	(a)	3.1	Visual
108	Any dud plugs missing	(a)	3.9.1	Visual”

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4.4.2.11: Add the following:

PARAGRAPH	TITLE			DRAWING NUMBER
"4.4.2.11	Dud Plug	SHEET 1 OF 1		12961001
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Push in - push out test	30	3.9.2	Test/4.5.4
Minor				
201	Poor workmanship	30	3.6	Visual"

4.4.3.1: Delete in its entirety and substitute:

"4.4.3.1 Proving ground test."

4.4.3.1.1, Add:

"The refurbished cartridges shall be observed for the following characteristics:

a. Misfires, major defect (see 3.6.1)
b. Duds, major defect (see 3.6.5)
c. Short rounds, major defect (see 3.6.2)
d. Premature burst, critical defect (see 3.6.6)
e. Metal parts separation, major defect (see 3.6.3)
f. Precision, major defect (see 3.6.4)
g. Weapon damage, major defect (see 3.6.7)
h. Improper impact signature, major defect (see 3.6.8)
i. All dud plug heads out of any one projectile in the inert fuze ballistic test, Critical defect (see 3.6.9.2)
j. Two or more dud plug heads remain in the live fuze ballistic test, Major defect (see 3.6.9.1)".

4.4.3.1.2, Add:

"i. All three dud plug heads (regardless of stems) are missing from the projectile body in the inert fuze ballistic test.

j. Two or more dud plug heads (regardless of stems) fail to exit projectile bodies (accumulatively, i.e., two remaining in a single projectile or one each remaining in two projectiles) in the live fuze ballistic tests (except in the case of a particular round that was an actual dud, in which case paragraph "i." above applies)."

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* 4.4.3.1.2(b), Add after first sentence:

"Graze impact duds shall not be counted for score. These shall be recorded for information only and shall be x-rayed to document that the fuze was armed. Side impact duds shall be recorded for information only and x-rayed to document that the fuze was armed. If more than four side impacts occur, the lot represented by the sample shall be referred to the technical agency for disposition. In either of the above cases, if the x-ray reveals that the fuze did not arm, the fuze shall be considered to be a dud."

* 4.4.3.1.2(c), Delete in its entirety and substitute:

"c. One or more short rounds occur at +110 degrees F or +70 degrees F.

Two or more short rounds occur at 0 degrees F."

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Add the following:

"d. One group of 24 each with inert fuzes conditioned for at least 24 hours at 0 degrees F +/- 2.5 degrees F (see 4.4.3.1.2), shall be fired at charge 3."

4.5.1.1, last paragraph: Delete and substitute:

"The individual deflections and ranges (accuracy of both to 0.01 meter) and muzzle velocities shall be calculated to the nearest 0.1 meter and recorded on the firing record for groups a, b, and c above. All rounds shall be fired onto a flat, nongrassy earthen surface that is not marshy or frozen."

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Add the following:

"4.5.4: Push in - push out test. Major defect. A hardened steel test plate of 1 x 2 x 1/4 inch is required for this test. There is to be a 0.250 ± .001 diameter thru hole (having an internal surface finish of 32 maximum) in the steel plate. The sample of dud plugs to be tested shall be preconditioned at a temperature of 100 degrees F ± 5 degrees F for a period of not less than 24 hours nor more than 48 hours prior to being subjected to this test. The preconditioned dud plugs shall be tested not earlier than 15 minutes after removal from the

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conditioning box nor later than one (1) hour after removal from the conditioning box. Press a sample dud plug into the hole using an arbor press until the head is 0.0625 ± 0.02 inch above flush with the plate surface. At this point, the push in test is completed. Record the maximum force. Turn the steel plate over and push the dud plug out of the steel plate to 0.0625 ± 0.02 inch above flush using the same arbor press. Record the maximum force required to push out the dud plug. At this point, the push out test is completed. Compute sample mean (xbar) and standard deviation (sd). Insert these into the following formulas:

Accept if:

a. xbar(In) = 37.0 pounds max., xbar(In) + 3sd(In) = 40.0 pounds max.

and

b. xbar(Out) = 30.0 pounds min., xbar(Out) - 3sd(Out) = 25.0 pounds min. and no individual x(Out) less than 25.0 pounds.

Any dud plug sample that does not meet the above requirements shall be cause for lot rejection.

Each test sample quantity may be examined for outliers (high side only) using the Grubbs T statistical test (see 6.12). A maximum of two outliers may be removed from the sample quantity."

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6.10 Delete in its entirety.

6.11 Add the following:

"6.11 Inspection lot definition. A homogeneous collection of units of product or one shipment of component parts or assemblies from a subcontractor, bearing identification and treated as a unique entity from which a sample is to be drawn and inspected (and tested) to determine conformance with the acceptability criteria."

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6.12 Add the following:

"6.12 Outlier test. When the accept criteria is not met and if it is suspected that a statistical outlier may be present in the sample of 30 observations. Arrange the observed values in order of increasing magnitude (X1, X2 .. Xn-1, Xn. Using the statistics calculated for the sample, calculate Tn where:

If X1, the smallest value, rather than Xn, is the doubtful value the test criterion is:

$$Tn = \frac{Xn - \bar{X}}{s}$$

If X1, the smallest value, rather than Xn, is the doubtful value the test criterion is:

$$T1 = \frac{\bar{X} - X1}{s}$$

Where:

$\bar{X}$ = arithmetic average of all n values
s = estimate of the population standard deviation based on the sample data calculated as follows:

$$s = \left[\frac{\sum (x_i - \bar{x})^2}{n-1}\right]^{1/2} = \left[\frac{n\sum x_i^2 - \left(\sum x_i\right)^2}{n(n-1)}\right]^{1/2}$$

If Tn is less than or equal to 2.745, the highest value may not be eliminated. The lot is rejected. If Tn is greater than or equal to 2.745, the largest value obtained in the sample shall be eliminated and the sample statistics X and s shall be recalculated using the twenty-nine remaining values. This shall be compared to the accept criteria. If the criteria is met, accept the lot. If not, the same procedure shall be followed for the Xn-1 observation. Calculate Tn-1 based on the new statistics of the remaining sample of 29. If Tn-1 is less than or equal to 2.730, the lot is rejected. If Tn-1 is greater than or equal to 2.730, then this observation shall be eliminated and the sample statistics calculated based on the remaining 28 observations. If these values meet the accept criteria, accept the lot. If the accept criteria is still not met, reject the lot."

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6.13: Add the following:

"6.13 Marshy ground. Marshy ground is defined as an earthen surface that contains standing water or free flowing (fluid) mud or a combination of both.

6.14 Add the following:

"6.14 Frozen ground. Frozen ground is defined as an earthen ground within which water has frozen creating a hard surface."

* Add the following paragraphs:

"6.15 Graze impact dud. A graze impact dud is a fuze dud caused solely by the fuze hitting an object in the impact area (e.g. rocks) on its side and damaging the fuze prior to ground impact. This shall be determined by the proving ground test director based on his examination of the round's point of impact.

6.16 Side impact dud. A side impact dud is a fuze dud caused solely by the cartridge impacting on its side (without the fuze hitting an object prior to ground impact where the fuze striker shows no evidence of impact. This shall be determined by the proving ground test director based on his examination of the rounds point of impact."

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